



Summary

This is an overview of approval 40584164 (Ask to Be Sure: Informative Interactions for Confident Multi-Turn LLM Recommendation for ICLR 2026). The approval is DRAFT.

Basic Details

Title

Ask to Be Sure: Informative Interactions for Confident Multi-Turn LLM Recommendation for ICLR 2026

Summary

[The authors are requesting VP-level pre-authorization for a scientific publication (journal article or conference presentation). The authors understand that even after this pre-authorization, they will need to provide evidence that they have received approvals for all datasets used in their research before they will be approved for final publication. The authors have been told that AFTER this ticket has their org VP approval, they still need to wait for the monthly VP approval meeting for their pre-auth approval to be complete.]

Key Details

Name	Value
What is the title of the paper?	Ask to Be Sure: Informative Interactions for Confident Multi-Turn LLM Recommendation
Did you review the new Scipub tenets for publishing? https://w.amazon.com/bin/view/Amazon-Science/publishing/Overview	Yes
Are you attaching a full/final version of the publication to this ticket?	No
Primary Question 1. Nontechnical Research Summary: Give a VERY simplified, SVP-level, summary of the research discussed in your publication.	Our research addresses the challenge of fine-tuning language models for Prime Video's AI assistant to provide accurate movie recommendations through strategic interactions and effective preference understanding and solicitation. This work is the first to use entropy reduction as a reward mechanism for fine-tuning conversational recommendation systems, eliminating the need for labeled "correct" recommendations during the training process while encouraging strategic interactions that solicit user preferences. This research provides a novel approach for improving AI assistant interactivity and recommendation accuracy, particularly when ground truth recommendations aren't available, enabling more practical deployment of conversational AI in real-world scenarios like Prime Video. If published, this work will establish a new paradigm for fine-tuning LLM-based conversational recommendation systems that could be adapted for recommendations in other domains.

Test 1a. Enter the sentence(s) from your summary that explains our use case for the research discussed in this publication.	Our research addresses the challenge of fine-tuning language models for Prime Video's AI assistant to provide accurate movie recommendations through strategic interactions and effective preference understanding and solicitation.
Test 1b. Enter the sentence(s) from your summary that explains why this research is novel.	This work is the first to use entropy reduction as a reward mechanism for fine-tuning conversational recommendation systems, eliminating the need for labeled "correct" recommendations during the training process while encouraging strategic interactions that solicit user preferences.
Test 1c. Enter the sentence(s) from your summary that explains how this research is contributing to critical work in our domain.	This research provides a novel approach for improving AI assistant interactivity and recommendation accuracy, particularly when ground truth recommendations aren't available, enabling more practical deployment of conversational AI in real-world scenarios like Prime Video.
Test 1d. Enter the sentence(s) from your summary that explains what impact this publication will have on the scientific field if it is published.	If published, this work will establish a new paradigm for fine-tuning LLM-based conversational recommendation systems that could be adapted for recommendations in other domains.
Primary Question 2. Risks of Publishing: Give an honest assessment of the risk to the business involved in publishing this research rather than keeping it to ourselves.	Based on the methodology we would publish, a peer organization would need approximately 6-12 months to implement and adapt our entropy-based reward approach, requiring significant engineering effort to build their own conversation simulation systems and fine-tuning infrastructure from scratch. We are NOT giving out a significant advantage because this research represents a preliminary proof of concept based on public datasets, and productionizing it would require much more effort such as a much larger recommendation ecosystem that includes proprietary user data, content catalogs, advanced language models, and existing recommendation infrastructure that competitors cannot replicate. We ARE holding back critical implementation details including our specific hyperparameter optimization strategies and Prime Video's existing AI assistant recommendation pipeline.
Test 2a. Enter the sentence(s) from #2 that explains how long it would take a peer to leverage this research based on what we publish.	Based on the methodology we would publish, a peer organization would need approximately 6-12 months to implement and adapt our entropy-based reward approach, requiring significant engineering effort to build their own conversation simulation systems and fine-tuning infrastructure from scratch.
Test 2b. Enter the sentence(s) from #2 that explains how we are NOT giving out a significant advantage.	We are NOT giving out a significant advantage because this research represents a preliminary proof of concept based on public datasets, and productionizing it would require much more effort such as a much larger recommendation ecosystem that includes proprietary user data, content catalogs, advanced language models, and existing recommendation infrastructure that competitors cannot replicate.
Test 2c. Enter the sentence(s) from #2 that explains what we ARE holding back to protect our IP and secret sauce. Remember tenet #2: Protect Amazon!	We ARE holding back critical implementation details including our specific hyperparameter optimization strategies and Prime Video's existing AI assistant recommendation pipeline.

Test 2d. Enter the sentence(s) from #2 that mentions other UP safeguards such as patents filed, if applicable.	N/A
Primary Question 3. Benefits of Publishing: Describe the business case for publishing this article.	Publishing this research establishes Amazon as a thought leader in conversational AI and recommendation systems, enhancing our reputation for innovation which helps attract top-tier research talent and strengthens partnerships with academic institutions. Additionally, engaging with the scientific community through publication provides valuable external validation of our approach and creates opportunities for collaborative research that can accelerate our internal development efforts. This publication will move the scientific community's research direction toward conversational recommendation problems that directly benefit us by encouraging researchers to focus on entropy-based reward mechanisms, conversation simulation techniques, and the challenge of training recommendation systems without ground truth labels - all areas where continued external research will generate insights and methodological improvements that we can leverage for Prime Video's AI assistant development.
Test 3a. Enter the sentence(s) from #3 that explains why is it beneficial to the business to publish this instead of just working internally.	Publishing this research establishes Amazon as a thought leader in conversational AI and recommendation systems, enhancing our reputation for innovation which helps attract top-tier research talent and strengthens partnerships with academic institutions. Additionally, engaging with the scientific community through publication provides valuable external validation of our approach and creates opportunities for collaborative research that can accelerate our internal development efforts.
Test 3b. Enter the sentence(s) from #3 that details how this publication will move the scientific community's research direction to work on problems that benefit us.	This publication will move the scientific community's research direction toward conversational recommendation problems that directly benefit us by encouraging researchers to focus on entropy-based reward mechanisms, conversation simulation techniques, and the challenge of training recommendation systems without ground truth labels - all areas where continued external research will generate insights and methodological improvements that we can leverage for Prime Video's AI assistant development.
Primary Question 4. Other Amazon Businesses: Would this publication have any impact (direct or indirect) on any other Amazon businesses (such as Devices, AWS, Stores, etc)?	None
Primary Question 5. Confidential Information: You should seek to reduce or eliminate the use of Amazon customer data and confidential information as much as possible.	The experiments are conducted on public datasets without using Amazon user data.
VENUE: Where and in what format do you want to publish?	International Conference on Learning Representations (ICLR) 2026 Conference Proceedings
VENUE: What tier venue is this and why	ICLR is a top-tier venue that ranks as one of the three most prestigious

was this venue strategically the best? Remember the tenets: We publish to be recognized leaders in the field and choose top venues and we optimize timing!	conferences in machine learning alongside NeurIPS and ICML. ICLR focuses specifically on representation learning and novel training methodologies, making it the perfect venue for our entropy-based reward approach. The conference values both theoretical contributions and practical applications, which matches our work perfectly. Publishing at ICLR establishes Amazon as a leader in the most competitive AI research venue. This gives us maximum visibility among researchers, universities, and competitors. And ICLR's timeline fits the progress of the internship.
APP DEADLINE: What is the deadline to apply for this publication?	Sat Sep 20 06:59:59 UTC 2025
CAMERA-READY DEADLINE: What is the last date you could withdraw or make changes to the publication if accepted?	Sun Mar 15 06:59:59 UTC 2026
AUTHORS: List All Amazonian Authors	Cedar Bai, Duanshun Li, Zhenyu Liao, Sheikh Sarwar, Huiyuan Chen, Yuan Chen, Changhe Yuan, Haiyang Zhang, Qilin Qi
AUTHORS: List all external collaborating authors	
Who is the L8 scientist or tech leader in your direct reporting line? (If you do not have an L8 scientist or tech leader in your direct line, recommend an L8 scientist or tech leader who is the closest fit.)	Qilin Qi
Who owns the IP associated with this publication? Link to any dataset approvals you have.	We use public datasets.
If this is written in collaboration with a university, link to the ticket for that contract or say N/A.	
If this is a collaboration with a university: what form of support did Amazonians provide to earn this authorship: mentorship or did you contribute to the writing?	
Did you file a patent disclosure? This is REQUIRED. If not explain why.	
Did you update the PV Publication Tracker Quip? https://quip-amazon.com/HbHIAAwZcYfn/PVS-Publication-Tracker	

Approval Information

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Requester Bai, Cedar

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Approval Plan

Level 1	Manager Level 1 REQUESTERS_MANAGER ANT Group (Lnull)
Level 2	L8 Approver Level 2
Level 3	VP Level 3
Level 4	Research PM Check Level 4 Strobach, Natalie Sr. Program Mgr-Research Prgms (L6)
Level 5	IP Legal Level 5 Friday, Jason Corporate Counsel, Patents (L6)

Approval General Viewers

[Medioni, Gerard](#) Distinguished Scientist

[Winston, BA](#) VP, Technology, Prime Video

[REQUESTERS_MANAGER](#) DYNAMIC_ACTOR

[REQUESTERS_SKIP_LEVEL_MANAGER](#) DYNAMIC_ACTOR

Approval Admins

[Wu, Yongjun](#) Sr. Principal SDE

[Strobach, Natalie](#) Sr. Program Mgr-Research Prgms

[Bhat, Vimal](#) Sr. Manager Applied Science

Questionnaire

Is this a request to approve a transaction with committed revenue or spend over \$100 million in value?

No

Approvers should be asked to "recommend" this approval.

No

Can a requestor approve their own request as a member of a group?

Yes

Should escalations be enabled for this request?

No